UNIT - I

Solution of Algebraic & Transcendental Equations

(Bisection, Iteration, Regula-Falsi, Newton-Raphson, Jacobi & Gauss-Seidel)

1. Introduction

In engineering we often need the value of x that satisfies $f(x)=0$. Such a value is called a ROOT (or ZERO) of the equation.

- *Algebraic equation: a polynomial in x. e.g. $x^3-2x-5=0$, $x^2-3x+2=0$.*
- *Transcendental equation: contains trigonometric, exponential or logarithmic terms. e.g. $x\,e^x=2$, $\cos x=3x-1$, $x\log_{10}x=1.2$.*
- *Exact (analytical) formulas exist only for degree ≤ 4. For others we use NUMERICAL methods, which give an approximate root by successive approximation.*

Basic facts to remember

- *Intermediate Value Theorem: if $f(x)$ is continuous on $[a,b]$ and $f(a)\cdot f(b)<0$, then at least one root lies in (a,b).*
- *Initial approximation is found by TRIAL: tabulate $f(x)$ for $x=0,1,2,\ldots$ and look for a sign change.*
- *Descartes rule: number of +ve roots of a polynomial $\le$ number of sign changes in $f(x)$.*

Order of convergence

If $e_n=x_n-\alpha$ is the error at the n-th step and $|e_{n+1}|\approx k|e_n|^p$ then p is the ORDER of convergence.

Method	Order p	Needs	Speed
Bisection	1 (linear)	$f(a)f(b)<0$	slow, sure
Iteration	1 (linear)	$\lvert\varphi'(x)\rvert<1$	slow
Regula-Falsi	1 (about 1.0)	$f(a)f(b)<0$	moderate
Newton-Raphson	2 (quadratic)	$f'(x)\neq 0$	very fast

Algorithm idea: start with a guess $\Rightarrow$ improve it by a formula $\Rightarrow$ repeat until two successive values agree to the required decimal places.

2. Bisection Method (Bolzano / Interval-halving)

Principle

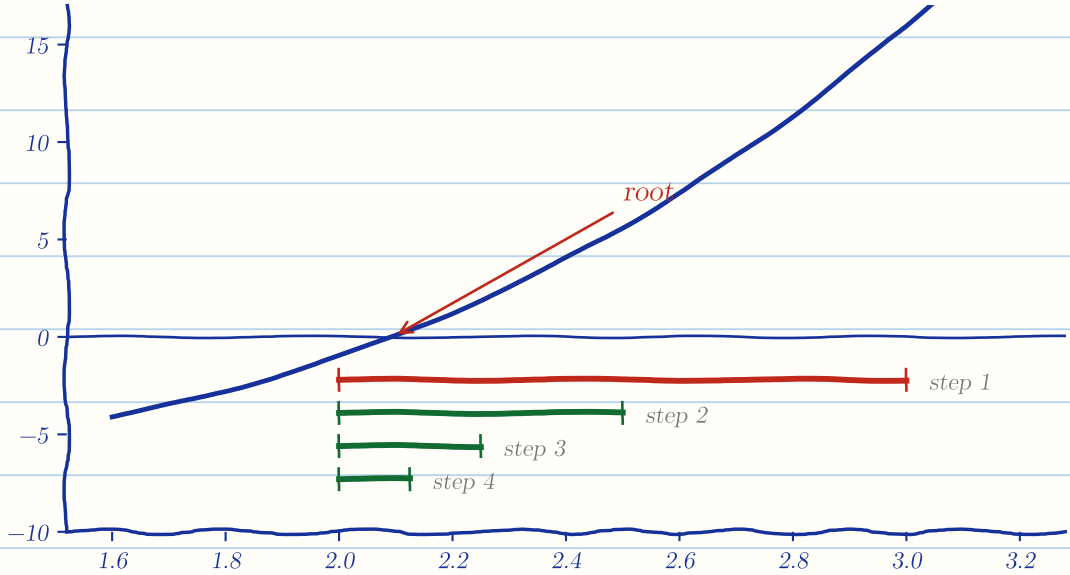
Let f be continuous on $[a,b]$ with $f(a)\,f(b)<0$. Then a root lies in (a,b). Bisect the interval at the mid-point:

$$x_0=\frac{a+b}{2}$$

- *If $f(x_0)=0$, then x_0 is the root.*
- *If $f(a)\,f(x_0)<0$, the root lies in (a,x_0), so set $b=x_0$.*
- *Otherwise the root lies in (x_0,b), so set $a=x_0$.*

Repeat. After every step the interval length is HALVED.

Bisection: interval halves each time



Number of iterations needed

After n steps the length is $(b-a)/2^n$. For accuracy ε :

$$\frac{b-a}{2^n} < \varepsilon \;\Rightarrow\; n \geq \frac{\log(b-a) - \log\varepsilon}{\log 2}$$

Merits: always converges, simple. Demerits: slow (linear), cannot find double roots / roots where f does not change sign.

Example 1. Find a real root of $x^3 - 2x - 5 = 0$ by bisection (3 decimals)

$f(2) = 8 - 4 - 5 = -1 < 0$ *and* $f(3) = 27 - 6 - 5 = 16 > 0$, *so a root lies in* $(2, 3)$.

n	a (−ve)	b (+ve)	$x_0 = (a+b)/2$	$f(x_0)$
1	2.0000	3.0000	2.5000	+5.6250
2	2.0000	2.5000	2.2500	+1.8906
3	2.0000	2.2500	2.1250	+0.3457
4	2.0000	2.1250	2.0625	-0.3513
5	2.0625	2.1250	2.0938	-0.0089
6	2.0938	2.1250	2.1094	+0.1668

n	a (−ve)	b (+ve)	$x_0 = (a+b)/2$	$f(x_0)$
7	2.0938	2.1094	2.1016	+0.0786
8	2.0938	2.1016	2.0977	+0.0347
9	2.0938	2.0977	2.0957	+0.0129
10	2.0938	2.0957	2.0947	+0.0020

After 10 steps $x \approx 2.0947$ *and* $f(x_0) \approx 0.002$, *so the root is* $x \approx 2.095$ *(true value 2.0946).*

Example 2. Find the positive root of $x^3 - x - 1 = 0$ in (1,2)

n	a	b	x_0	$f(x_0)$
1	1.0000	2.0000	1.5000	+0.8750
2	1.0000	1.5000	1.2500	-0.2969
3	1.2500	1.5000	1.3750	+0.2246
4	1.2500	1.3750	1.3125	-0.0515
5	1.3125	1.3750	1.3438	+0.0826
6	1.3125	1.3438	1.3281	+0.0146
7	1.3125	1.3281	1.3203	-0.0187

Root ≈ 1.32 *(exact root 1.3247). Interval length after 7 steps = 1/128 = 0.0078.*

3. Iterative Method (Fixed-point)

Rewrite $f(x) = 0$ *in the form* $x = \varphi(x)$. *Start with* x_0 *and compute*

$$x_{n+1} = \varphi(x_n), \quad n = 0, 1, 2, \ldots$$

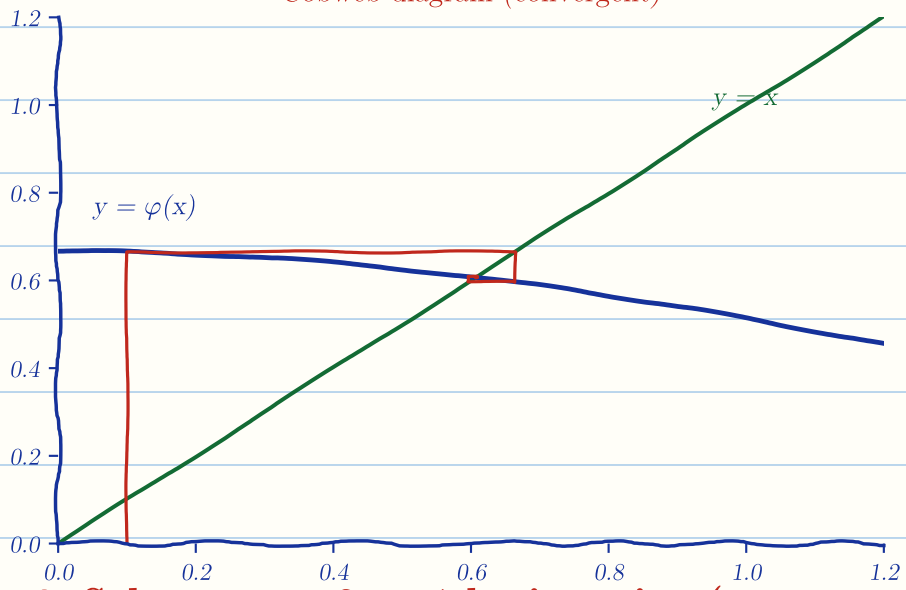
Key condition

Condition for convergence: $|\varphi'(x)| < 1$ *for all* x *in the interval containing the root.*

Smaller $|\varphi'| \Rightarrow$ *faster convergence. Error:* $e_{n+1} \approx \varphi'(\alpha)\, e_n$ *(linear).*

Choosing φ*: write the equation in several ways, pick the one with* $|\varphi'(x)| < 1$ *near the root. e.g. for* $x^3 + x - 1 = 0$ *take* $x = (1+x)^{-1/2}$ *or* $x = 1 - x^3$ *(check derivative!).*

Cobweb diagram (convergent)



Example 3. Solve $\cos x = 3x - 1$ **by iteration (start** $x_0 = 0.5$**)**

Write $x = \varphi(x) = \dfrac{1+\cos x}{3}$, $\varphi'(x) = -\dfrac{\sin x}{3}$, $|\varphi'| \le \frac{1}{3} < 1 \Rightarrow$ *convergent.*

n	x_n	$x_{n+1} = (1+\cos x_n)/3$
0	0.50000	0.62586
1	0.62586	0.60349
2	0.60349	0.60779
3	0.60779	0.60697
4	0.60697	0.60713

n	x_n	$x_{n+1} = (1+\cos x_n)/3$
5	0.60713	0.60710
6	0.60710	0.60710

Root ≈ 0.60710 *(correct to 4 decimals; cos taken in RADIANS).*

Example 4. Find a root of $2x - \log_{10} x = 7$ (near 3.8)

$x = \varphi(x) = \frac{1}{2}(\log_{10} x + 7)$, $\varphi'(x) = \frac{0.4343}{2x}$ *which is* < 1 *for* $x \approx 3.8$.

n	x_n	x_{n+1}
0	3.80000	3.78989
1	3.78989	3.78931
2	3.78931	3.78928
3	3.78928	3.78928
4	3.78928	3.78928

Hence the root is $x \approx 3.78893$.

4. Regula-Falsi Method (Method of False Position)

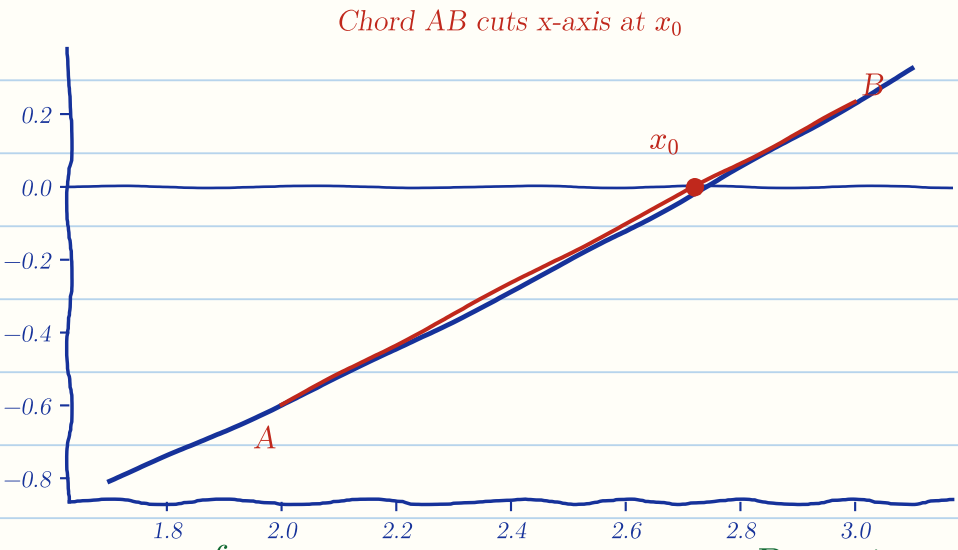
Replace the curve $y = f(x)$ *on* $[a, b]$ *by the CHORD joining* $A(a, f(a))$ *and* $B(b, f(b))$ *. The point where the chord cuts the* x*-axis is the next approximation.*

Equation of chord: $y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$*. Put* $y = 0$*:*

Regula-Falsi formula

$$x_0 = \frac{a\,f(b) - b\,f(a)}{f(b) - f(a)} = a - \frac{f(a)}{f(b) - f(a)}\,(b - a)$$

- *If* $f(a)f(x_0) < 0$ *the root is in* (a, x_0)*: replace* b *by* x_0*. Otherwise replace* a *by* x_0*.*
- *Repeat, always using the NEW end-points in the formula, until successive values agree.*



Merits: converges for sure, no derivative needed. Demerit: one end-point may stay fixed, so convergence is slow (order ≈ 1).

Example 5. Find a root of $x\log_{10} x = 1.2$ by Regula-Falsi

$f(2) = -0.5979 < 0,\ f(3) = +0.2314 > 0$, *root in* $(2,3)$. *With* $a = 2,\ b = 3$:

n	a	b	x_0	$f(x_0)$
1	2.0000	3.0000	2.72101	-0.01709
2	2.7210	3.0000	2.74021	-0.00038
3	2.7402	3.0000	2.74064	-0.00001
4	2.7406	3.0000	2.74065	-0.00000
5	2.7406	3.0000	2.74065	-0.00000

Root ≈ 2.7406 *(correct to 4 decimals).*

Example 6. Find the root of $x^3 - 2x - 5 = 0$ by Regula-Falsi (in (2,3))

n	a	b	x_0	$f(x_0)$
1	2.0000	3.0000	2.05882	-0.39080
2	2.0588	3.0000	2.08126	-0.14720

n	a	b	x_0	$f(x_0)$
3	2.0813	3.0000	2.08964	-0.05468
4	2.0896	3.0000	2.09274	-0.02020
5	2.0927	3.0000	2.09388	-0.00745

Compare: bisection took 10 steps for 3 decimals; here the root 2.0946 is reached much sooner.

5. Newton-Raphson Method

Let x_0 be an approximate root and $x_1 = x_0 + h$ the better value, so $f(x_0 + h) \approx 0$. Taylor series (first two terms):

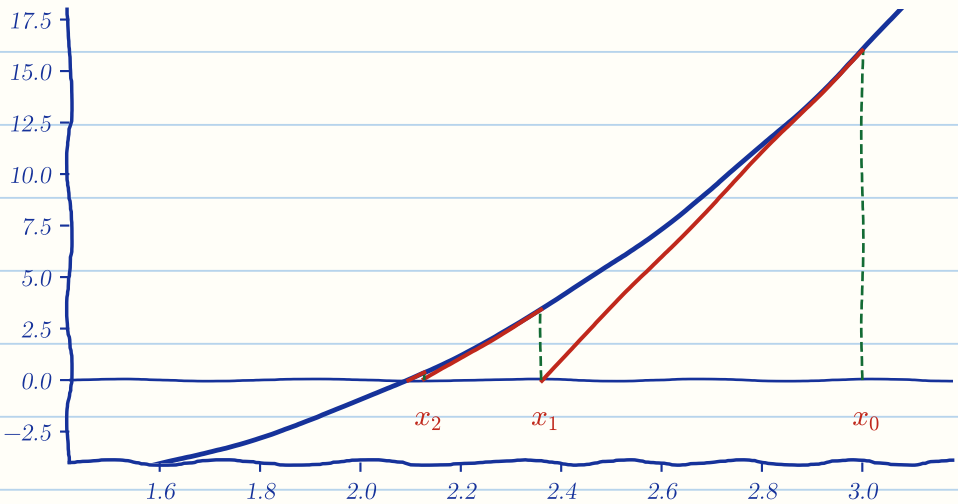
$$f(x_0) + h\, f'(x_0) = 0 \;\Rightarrow\; h = -\frac{f(x_0)}{f'(x_0)}$$

Newton-Raphson formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \qquad n = 0, 1, 2, \ldots$$

Geometrically: draw the TANGENT at $(x_n, f(x_n))$; the point where it meets the x-axis is x_{n+1}.

Tangents converge quickly to the root



Convergence & conditions

- *Order of convergence = 2 (quadratic): the number of correct decimals roughly DOUBLES every step.*

- *Convergence is guaranteed if* $|f(x)\,f''(x)| < |f'(x)|^2$ *in the interval around the root.*
- *Fails (or is slow) when* $f'(x_n) \approx 0$*; choose* x_0 *close to the root. Equivalent to iteration with* $\varphi(x) = x - f/f'$.

Example 7. Find a root of $x^3 - 2x - 5 = 0$ (take $x_0 = 2$)

$f(x) = x^3 - 2x - 5, f'(x) = 3x^2 - 2.$

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	2.00000	-1.00000	10.00000	2.10000
1	2.10000	0.06100	11.23000	2.09457
2	2.09457	0.00019	11.16165	2.09455
3	2.09455	0.00000	11.16144	2.09455

Root ≈ 2.09455. *Only 3 iterations needed!*

Example 8. Find a real root of $x\,e^x = 2$ (take $x_0 = 1$)

$f(x) = xe^x - 2,\ \ f'(x) = e^x(1 + x).$

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	1.00000	0.71828	5.43656	0.86788
1	0.86788	0.06716	4.44902	0.85278
2	0.85278	0.00077	4.34694	0.85261
3	0.85261	0.00000	4.34575	0.85261

Root ≈ 0.85261.

Important deductions from Newton-Raphson formula

(i) Square root of N $(f = x^2 - N)$*:*

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$$

(ii) Reciprocal of N $(f = \frac{1}{x} - N)$*:* $x_{n+1} = x_n\,(2 - N x_n)$

(iii) k-th root of N $(f = x^k - N)$: $x_{n+1} = \frac{1}{k}\left[(k-1)x_n + \frac{N}{x_n^{k-1}}\right]$

Example 9. Find $\sqrt{12}$ using Newton-Raphson ($x_0 = 3.5$)

n	x_n	$x_{n+1} = \frac{1}{2}(x_n + 12/x_n)$
0	3.50000	3.46429
1	3.46429	3.46410
2	3.46410	3.46410

$\sqrt{12} = 3.46410$. *Check:* $3.4641^2 = 11.99999$.

Example 10. Find the reciprocal of 18 ($x_0 = 0.05$)

n	x_n	$x_{n+1} = x_n(2 - 18x_n)$
0	0.050000	0.055000
1	0.055000	0.055550
2	0.055550	0.055556

$1/18 = 0.055555..$ *correct.*

6. System of Linear Algebraic Equations: Jacobi & Gauss-Seidel

For large systems direct methods (Cramer, Gauss elimination) are lengthy; ITERATIVE methods are used. Consider

$$a_1x + b_1y + c_1z = d_1, \quad a_2x + b_2y + c_2z = d_2, \quad a_3x + b_3y + c_3z = d_3$$

Convergence condition

Condition: the system must be DIAGONALLY DOMINANT, i.e.

$$|a_1| > |b_1| + |c_1|, \quad |b_2| > |a_2| + |c_2|, \quad |c_3| > |a_3| + |b_3|$$

If not, interchange the equations first to make it so.

Solve for the diagonal unknowns:

$$x=\frac{1}{a_1}(d_1-b_1y-c_1z),\quad y=\frac{1}{b_2}(d_2-a_2x-c_2z),\quad z=\frac{1}{c_3}(d_3-a_3x-b_3y)$$

(a) Jacobi iteration (method of simultaneous displacement)

Start with $x^{(0)}=y^{(0)}=z^{(0)}=0$ *(or any guess). Compute ALL new values using only the OLD values:*

$$x^{(k+1)}=\frac{1}{a_1}(d_1-b_1y^{(k)}-c_1z^{(k)}),\quad y^{(k+1)}=\frac{1}{b_2}(d_2-a_2x^{(k)}-c_2z^{(k)})$$

$$z^{(k+1)}=\frac{1}{c_3}(d_3-a_3x^{(k)}-b_3y^{(k)})$$

(b) Gauss-Seidel iteration (method of successive displacement)

Use the LATEST available value immediately in the next equation:

$$x^{(k+1)}=\frac{1}{a_1}(d_1-b_1y^{(k)}-c_1z^{(k)})$$

$$y^{(k+1)}=\frac{1}{b_2}(d_2-a_2x^{(k+1)}-c_2z^{(k)}),\quad z^{(k+1)}=\frac{1}{c_3}(d_3-a_3x^{(k+1)}-b_3y^{(k+1)})$$

Gauss-Seidel converges about TWICE as fast as Jacobi and needs less storage.

Example 11. Solve:

$10x+y+z=12,\quad 2x+10y+z=13,\quad 2x+2y+10z=14$

Diagonally dominant (\$10>1+1,\ 10>2+1,\ 10>2+2\$). Rewrite:

$x=\frac{1}{10}(12-y-z),\ y=\frac{1}{10}(13-2x-z),\ z=\frac{1}{10}(14-2x-2y).$

Jacobi (start 0,0,0):

Iteration	x	y	z
1	1.2000	1.3000	1.4000
2	0.9300	0.9200	0.9000

Iteration	x	y	z
3	1.0180	1.0240	1.0300
4	0.9946	0.9934	0.9916
5	1.0015	1.0019	1.0024
6	0.9996	0.9995	0.9993

Gauss-Seidel (start 0,0,0):

Iteration	x	y	z
1	1.2000	1.0600	0.9480
2	0.9992	1.0054	0.9991
3	0.9996	1.0002	1.0001
4	1.0000	1.0000	1.0000
5	1.0000	1.0000	1.0000

Solution: $x=1,\ y=1,\ z=1$. *Gauss-Seidel is correct to 4 decimals by the 5th step; Jacobi needs a few more.*

Example 12. Solve by Gauss-Seidel:

$20x+y-2z=17,\ 3x+20y-z=-18,\ 2x-3y+20z=25$

$x=\frac{1}{20}(17-y+2z),\ y=\frac{1}{20}(-18-3x+z),\ z=\frac{1}{20}(25-2x+3y).$

Iteration	x	y	z
1	0.8500	-1.0275	1.0109
2	1.0025	-0.9998	0.9998
3	1.0000	-1.0000	1.0000
4	1.0000	-1.0000	1.0000

Solution $x=1,\ y=-1,\ z=1.$

7. Quick Revision Sheet

Formulas at a glance

Bisection : $x_0 = \frac{a+b}{2}$; $n \geq \frac{\log(b-a) - \log \varepsilon}{\log 2}$

Regula-Falsi: $x_0 = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$ Newton: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

Iteration : $x_{n+1} = \varphi(x_n)$, *needs* $|\varphi'(x)| < 1$

Jacobi: use old values ; Gauss-Seidel: use newest values (faster).

Both need diagonal dominance.

- *Always check the sign change* $f(a)f(b) < 0$ *first and write* $f(a), f(b)$ *explicitly in the exam.*
- *Keep one extra decimal in intermediate steps; stop when two consecutive iterates agree to the asked decimals.*
- *Calculator in RADIAN mode for trigonometric equations.*
- *Expected exam questions: bisection/regula-falsi/N-R problem (5 marks each), Jacobi or Gauss-Seidel problem (5 marks), derivation of N-R formula (short answer).*